

SDS-PAGE Protein Separation

Laboratory Report — Molecular Weight Estimation by Standard Curve

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Technique category: Protein Biochemistry

1. Objective

To separate proteins according to molecular weight using SDS-PAGE, and to estimate the molecular weight of unknown protein bands by comparison with a protein ladder of known molecular weights.

2. Introduction

SDS-PAGE (Sodium Dodecyl Sulfate–Polyacrylamide Gel Electrophoresis) separates proteins by molecular size rather than native charge or shape. SDS, an anionic detergent, denatures proteins and coats them with a uniform negative charge proportional to mass, masking their intrinsic charge. Under an applied electric field, the polyacrylamide gel acts as a molecular sieve: smaller proteins migrate further than larger ones. For a set of ladder standards of known molecular weight, migration distance is linearly related to $\log_{10}(\text{molecular weight})$ — this relationship forms a standard curve that can be used to estimate the size of unknown protein bands.

3. Materials

- Protein sample(s) and pre-stained molecular weight marker (ladder)
- 30% acrylamide/bis-acrylamide, resolving and stacking gel buffers
- 10% SDS, 10% ammonium persulfate (APS), TEMED
- Laemmli sample buffer with reducing agent (β -mercaptoethanol or DTT)
- Tris-glycine-SDS running buffer
- Coomassie Brilliant Blue stain and destaining solution
- Vertical electrophoresis apparatus and power supply

4. Method

- **Gel casting:** Resolving gel (10–12%) cast and overlaid with a stacking gel (4–5%).
- **Sample preparation:** Protein mixed with Laemmli buffer, heated at 95–100°C for 5 minutes.
- **Loading:** Samples and ladder loaded into separate wells.
- **Electrophoresis:** Run at ~80V through the stacking gel, then ~120–150V through the resolving gel.
- **Staining:** Gel stained with Coomassie Brilliant Blue (30–60 min).
- **Destaining:** Gel washed until protein bands are clearly visible.
- **Analysis:** Migration distances measured and plotted against $\log(\text{molecular weight})$ of the ladder.

5. Results

Migration distances were recorded for eight ladder bands (15–180 kDa) and two unknown sample bands. A linear regression of $\log_{10}(\text{molecular weight})$ against migration distance produced the standard curve: **$\log_{10}(\text{MW}) = -0.0152 \times \text{distance} + 2.392$** (R-squared = 0.994), indicating a strong linear fit across the ladder range.

Band	Migration (mm)	Estimated MW (kDa)
Unknown_1	38.5	64.2
Unknown_2	52.1	39.9

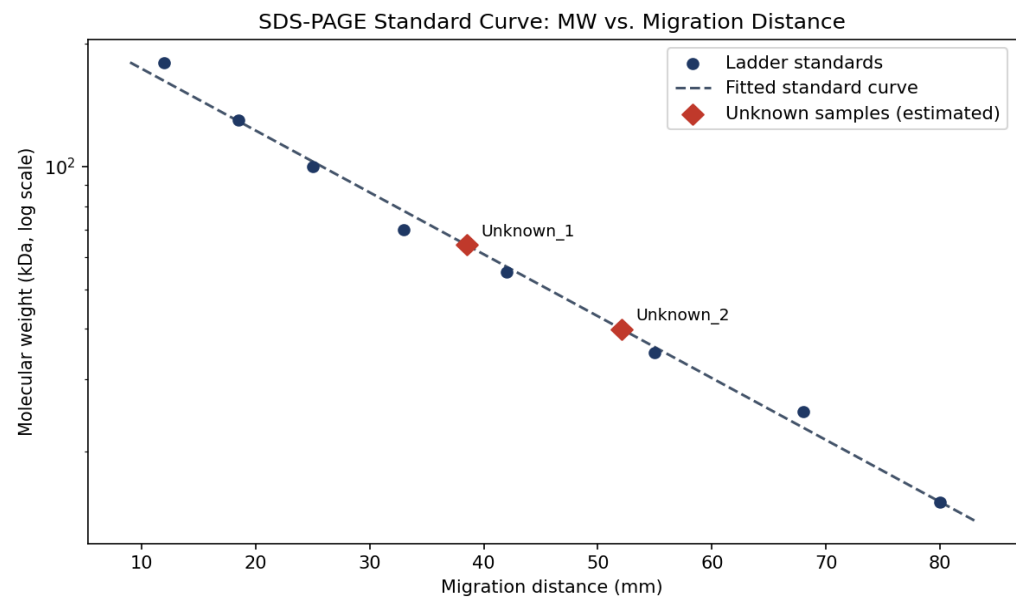


Figure 1. Standard curve (log-scale) fitted from ladder migration distances, with unknown bands plotted against it.

Both unknown bands fell within the migration range of the ladder standards, meaning their estimated molecular weights (64.2 kDa and 39.9 kDa) are interpolated rather than extrapolated — the more reliable case for standard-curve estimation.

6. Discussion

The high R-squared value (0.994) indicates the ladder bands migrated consistently with the expected inverse log-linear relationship between size and migration distance, supporting confidence in the fitted curve. Because both unknown bands fell inside the ladder's size range, their molecular weight estimates required interpolation only, which is generally more reliable than extrapolating beyond the smallest or largest ladder standard. In a real experimental setting, further confirmation (e.g. mass spectrometry or western blotting with a specific antibody) would be used to verify protein identity, since SDS-PAGE alone identifies size, not identity.

7. Conclusion

This exercise demonstrated the complete SDS-PAGE workflow, from gel preparation through protein separation to quantitative molecular weight estimation using a standard curve — a foundational technique for characterising protein samples in a biochemistry laboratory setting.